

# APS360: Sequence to Fitness Prediction

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GitHub: [https://github.com/Arthurzhou09/aps360\\_project.git](https://github.com/Arthurzhou09/aps360_project.git)

## Introduction

Protein sequences encode function. Alterations at certain sequence sites cannot be tolerated, whereas mutations at other sites can enhance protein activity. Therefore, characterizing mutation effects is important for designing variants with novel or enhanced functions. This project focuses on TEM-1  $\beta$ -lactamase, an antibiotic resistance enzyme produced in pathogenic bacteria. Novel  $\beta$ -lactam antibiotics have consistently been followed by modified  $\beta$ -lactamases that confer resistance to these antibiotics [1]. Therefore, accurate prediction of mutation effect on protein fitness can forecast evolution and guide the design of antibiotic resistant drugs. However, navigating the protein fitness landscape is difficult due to complex biophysical interactions resulting from backbone sterics and amino-acid interactions. Recent advances in high-throughput technologies such as deep mutational scanning (DMS) have generated large-scale datasets linking sequence variants to functional measurements. While this has improved mutation assessment, experimentally scaling to all higher order variants remains prohibitively expensive. To address this limitation, this project uses a deep learning approach to predict variant fitness from various biological signals that impact protein fitness (DMS data, Amino Acid (AA) properties, and protein structure) 1. **The goal of this project is to evaluate models trained on single mutations to predict higher-order mutation effects on TEM-1  $\beta$ -lactamase fitness.**

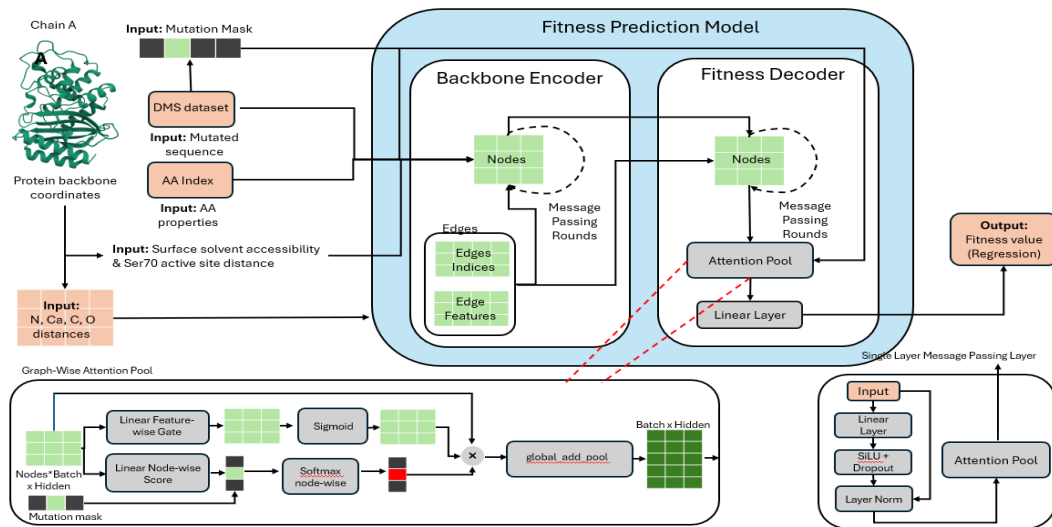


Figure 1: Inputs and outputs to the primary message passing network (GNN).

## Background & Related Work

**Statistical Models.** *Chen et al.* conducted a family-wide analysis of B1 metallo-beta-lactamases, showing that co-evolutionary trends can encode epistatic relationships [2]. *Huang et al.* introduced a computational model (LINSIGHT) that uses human genetic polymorphism data across closely related species to predict non-coding sites where mutations are likely to reduce fitness [3].

**Protein Language Models.** ProMEP is a self-supervised deep representation learning model that learns protein sequence and structural representations to predict both single- and multi-mutation effects, with reduced performance on higher-order mutants [4].

**Supervised Deep Learning.** Graph neural networks (GNN) are commonly used. *Huynh et al.* incorporate a physics-based metric, the Asymmetric Dynamic Coupling Index (ADCI), into training a graph convolution network (GCN), motivated by the idea that protein dynamics capture functional effects not explained by static structure alone [5]. *Zhang et al.* train a graph attention network (GAT) for missense variant prediction, constructing a graph per variant with the reference amino acid as the central node and neighboring amino acids as context nodes [6]. Edge features encode co-evolutionary strength, which can correlate with function and spatial proximity in the folded structure.

## Data Processing

**Dataset.** Single mutant [7] and consecutive double mutant [8] TEM-1  $\beta$ -Lactamase (*Escherichia coli*) experimental DMS data containing fitness values (normalized so that 0–1 indicates deleterious variants, 1 corresponds to wild type (WT), and values above 1 indicate beneficial variants), estimated fitness error, and mutation codes were collected. Rows with missing values or WT-to-WT mutations were discarded and a mutation code (*WT\_AMBLER\_MUT*) was created. WT residues were sequentially combined by Ambler position to obtain the experimental WT sequence. TEM-1  $\beta$  structural information was obtained from the *mmCIF* protein structure file (*PDB* = *1BTL*). Global sequence alignment (BioPython) mapped experimental mutation positions onto backbone sequence positions, removing the N-terminal signal peptide absent from the mature enzyme structure. The final processed dataset contains **5032** single and **11554** double variants (**n = 16586**) across 263 mutation sites. The most common mutation was Glycine and mutations occurred relatively equally across all 263 positions <sup>2</sup>. This report uses **single mutation data only**, where most variants were non-beneficial (**fitness < 1.0, 78%**). A PCA-compressed representation of 566 AAindex properties from [9] yielded 19 features that preserve 100% of the variance while reducing redundancy <sup>3</sup>. AAindex encoding captures physiochemical and biochemical properties such as hydrophobicity and polarity [10].

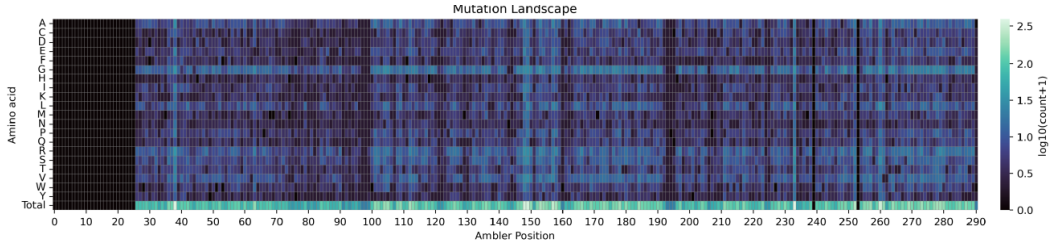


Figure 2: Single and pair AA mutation heatmap across the backbone sequence.



Figure 3: **A.** Node and edge features for main GNN. **B.** Baseline MLP.

**Feature Processing.** The mutated sequence, structural information, and AA properties were transformed into graph-based features for input into a **message passing network** 1. Each residue was represented as a backbone node, with edges generated from the union of spatial neighbours based on  $C\alpha$  distances and Dynamic-coupling index (DCI) [5]. DCI measures how residue fluctuations respond to perturbations at other sites by applying random forces to each  $C\alpha$  node in an elastic network model and normalizing the response [11]. Because this response is not always reciprocal, the asymmetric term  $DCI_{ij} - DCI_{ji}$  provides a directed measure of allosteric influence and is incorporated as an edge feature and used to generate directed graph edges. Spatial distances were

encoded using a Gaussian radial basis function to encode smooth distance embeddings. Each node contains a one-hot encoded AA identity, AA physicochemical properties, solvent exposed surface area calculated using the Shrake-Rupley algorithm in Biopython (probe radius = 1.40Å), and  $C\alpha$  distance to the Ser70 active-site residue. Additional node features include DCI direction, DCI pair, neighbour pair flags, and a binary mutation mask 13. The **baseline model** used flattened non-distance node features as input.

## Models and Training

Both models were trained and tested on the single DMS dataset using an 8:1:1 split. Models were trained with AdamW (weight decay  $1 \times 10^{-5}$ ), batch size 32, mean-squared-error loss, and 40 epochs with early stopping and reduce-on-plateau schedulers 4.

**Data split.** Data was split by sequence position such that each split contained unique mutation positions. **This more difficult split** tests generalization to unseen residues compared to random position splitting 4, which performs better but is less rigorous. Sequence mutation information was provided only through graph construction and backbone-derived node features. Fitness labels were z-score normalized using the training set and converted back to the original range during inference.

**GNN Hyperparameters.** Hyperparameters were selected by random search over seven fields: learning rate  $[5 \times 10^{-5}, 10^{-3}]$ , weight decay  $[10^{-5}, 10^{-1}]$ , dropout  $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ , hidden width  $\{32, 64, 96\}$ , message-MLP depth  $\{2, 3\}$ , encoder rounds  $\{1, 2, 3\}$ , and decoder rounds  $\{1, 2\}$ . Twelve configurations were ranked using validation Spearman coefficient due to the sparse high-fitness tail of MSE. Since the validation split contained only 27 residues (487 variants), the top three configurations were retrained across three seeds for stability and ranked by mean performance. The final model used learning rate  $5.3 \times 10^{-4}$ , weight decay  $9.1 \times 10^{-5}$ , dropout 0.5, hidden width 32, two-layer message MLPs, three encoder rounds, and one decoder round, with 236K parameters.

## Model Results

The **baseline model 3** achieved an MSE of  $0.18 \pm 0.25$ , Pearson  $r$  of 0.36, and Spearman  $\rho$  of 0.41 on the single mutation test split. The **message passing network 1** achieved an MSE of  $0.13 \pm 0.20$ , Pearson  $r$  of 0.58, and Spearman  $\rho$  of 0.63. The **random split performance** is included to show that higher performance can be achieved with an easier split that leaks data.4.

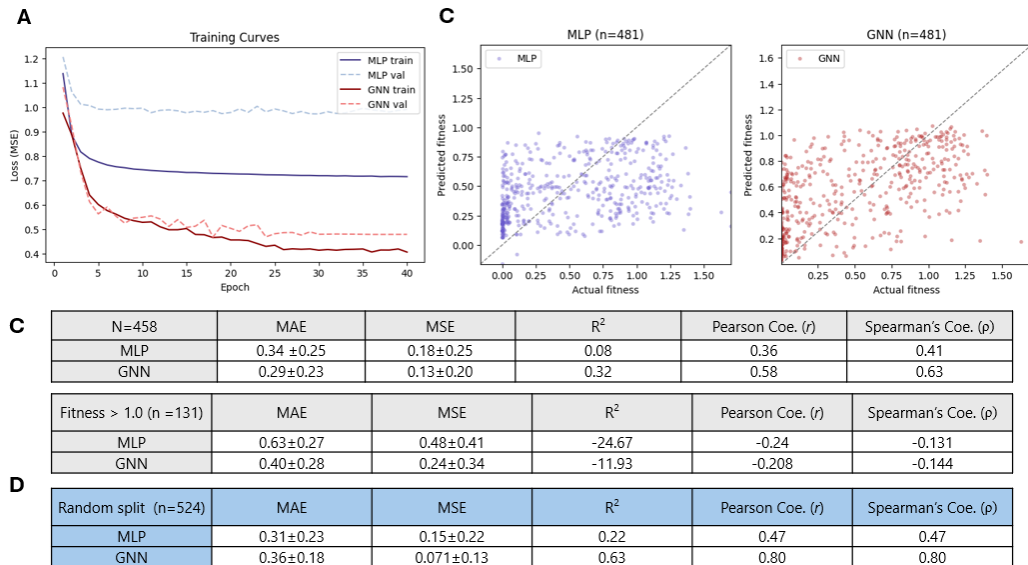


Figure 4: **A.** Structure split training curves. **B.** Predictions on structure split. **C.** Quantitative performance for structure split on entire set and underrepresented beneficial mutation subset. **D.** Performance on random split.

**Evaluation on New Data.** Double mutation DMS data is split to include only unseen training residues (n=507). In contrast to the single mutation DMS test set, the GNN's performance was

significantly better than the MLP. The GNN achieves a Pearson  $r$  of 0.70, Spearman  $\rho$  of 0.73 and comparable MAE and MSE, outperforming itself on unseen double mutant sequences despite having a similar fitness class distribution to the single mutant set 5.

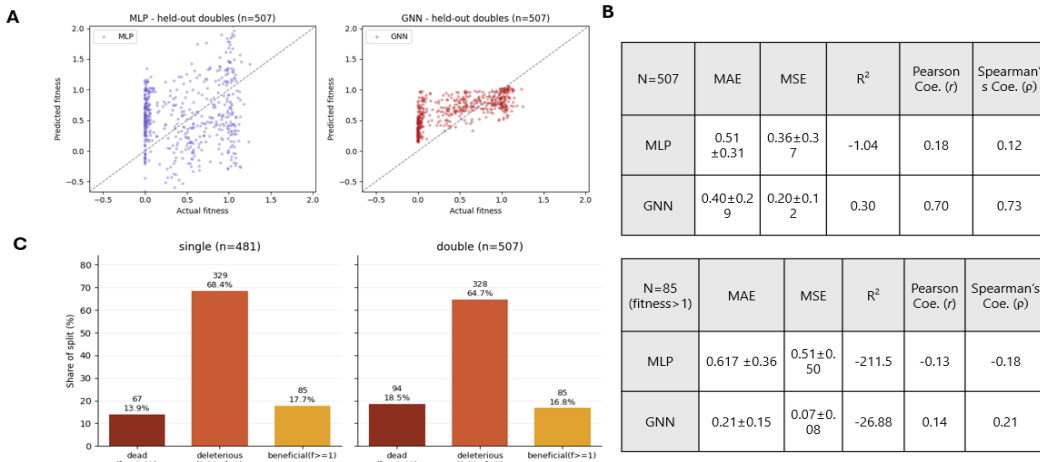


Figure 5: **A.** Predictions on unseen double mutants. **B.** Quantitative performance. **C.** Test split distribution of single vs double mutants

## Discussion

For both models, evaluation on the subset of beneficial single mutations (fitness > 1.0) from the test split 4 resulted in Pearson  $r$  and Spearman's  $\rho$  near zero. This suggests that the models struggle to learn sequence-function relationships underlying gain-of-function mutations. One explanation is that, because the dataset split is performed by residue position (all mutations at a given position fall within the same split), functionally critical positions with large fitness effects may be entirely absent from training, leaving the model without examples to generalize from. Another likely contributor is the scarcity of beneficial mutations, which may cause both models to oversmooth predictions toward the dominant near-zero fitness mode rather than capture the sparse high-fitness signal. **However, fitness signals are partially recovered by the GNN but not the MLP when evaluated on unseen double mutants.** Double mutants in this dataset always occupy adjacent backbone positions, meaning the substituted residues are directly connected in the structural graph and exchange messages from the first encoder round. In contrast, the MLP receives the two sites as independent features without an interaction term. Since the decoder also attention-pools over the mutated residues, a double mutant contributes two context-conditioned embeddings each mixed with its neighbour's state to the pooled representation rather than one. This suggests that structural context becomes more informative beyond single sites, **even when the model is not trained on higher-order mutations.** Additional training on higher-order mutational data may further improve performance, as structural information from multi-mutations may provide functional information unavailable from single mutations alone.

**Difficulty.** An attempt was made at self-supervised masked-residue pretraining on homolog/MSA data, scoring zero-shot mutation effects as  $\log P(\text{mutant}) - \log P(\text{WT})$  at masked sites. The first attempt pretrained a 938k-parameter GNN and scored below a rule based evolutionary model (zero-shot Spearman 0.08–0.16 vs. 0.21) on single DMS mutants. Using TEM-1  $\beta$ lactamase family-specific MSA increased zero-shot Spearman to 0.50, but increased the computation required. A single structure GNN is likely a poor choice for this task, which often requires large protein-language models. For reference, SOTA supervised fitness prediction models achieve average Spearman scores of 0.54 - 0.65 (e.g. Kermut [12], ESM-1v Embeddings [13], ProteinNPT [14]). SOTA Zero-shot models achieve Spearman scores of 0.50 (e.g. ProSST [15]).

## Ethical Considerations

Experimental fitness measures often rely on surrogate assays (e.g., antibiotic survival) that may not reflect true evolutionary fitness, limiting transferability to more complex systems. Data collection involving clinically relevant genes raises concerns about data sharing and misuse, and such models could potentially be misused to design variants with increased virulence.

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